# Fuzzing the Solidity Compiler

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# whoami

- Security engineer, Solidity team
- Semantic testing of Solidity compiler

Find security-critical bugs in the compiler before it is shipped

# tl;dr:

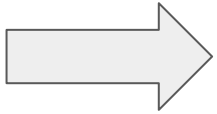
- Threat model: Incorrect code generation
- Randomly generated valid Solidity programs test compiler
- Found 2 security relevant bugs in EVM optimizer
  - Low or very low security impact
- Found 5 other bugs in experimental optimizer
- Continuous fuzzing for early bug discovery

# Introduction

# Threat model

- Compiler user (developer) is not malicious
- Bugs introduced by the optimizer

```
function foo() -> x {
  x := 2
}
mstore(0, foo())
```



```
mstore(0, 2)
```

# Fuzz testing in a nutshell

```
while not ctrl + c
do
   input=gen_input()
   runProgram(input)
done
```

# Limitation of random fuzzing

```
contract C {
  function foo()
public {

do_something();
  }
}
```

Mutation

```
contract C {
  fu#!3ion foo()
puX^&c {

do_something();
  }
}
```

Accepted by parser

Rejected by parser

Fuzzing a compiler requires generating valid programs...

... generating a valid program requires structure awareness

# Approach

# Write a specification

Specification written in protobuf language

```
message Block {
  Repeated Statement stmts;
}
...
message program {
  repeated Block blocks;
}
```

Full spec:
https://github.com/ethereum/solidity/blob/develop/test/tools/ossfuzz/yulProto.proto

# Input generation

- Input generated and mutated by libprotobuf-mutator
- Each input is a tree

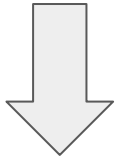
```
blocks { stmts { ifstmt { condition {
binaryOp { eq { op1: varref{id: 0} op2: 0}
} } } } }
```

# Input conversion

- Converter is source-to-source translator
- Input: protobuf serialization format
- Output: yul program

# Example

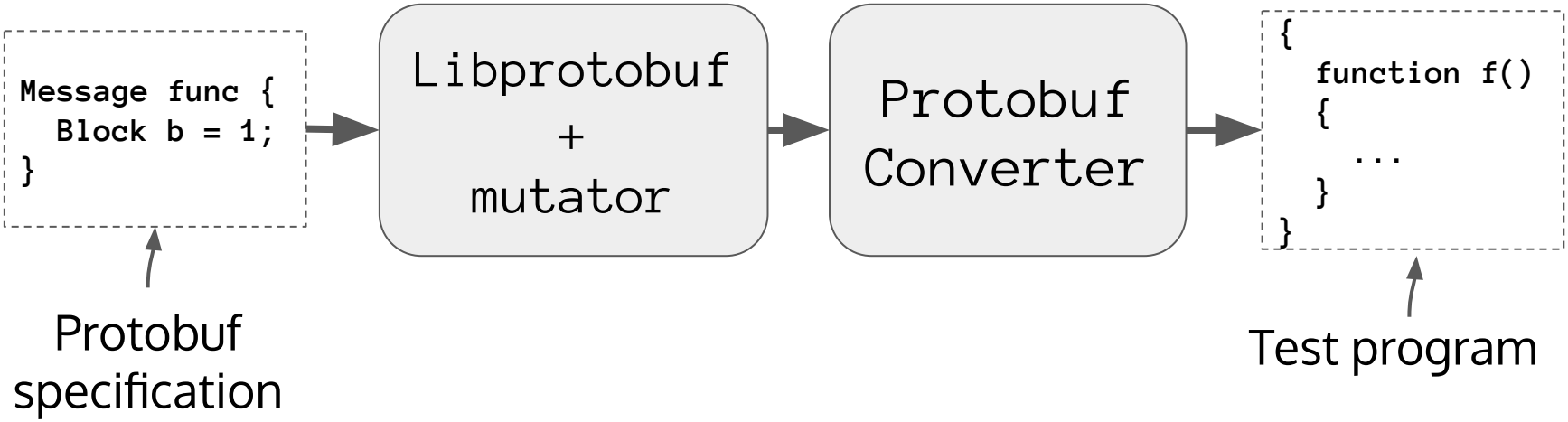
```
blocks { stmts { ifstmt { condition {
binaryOp { eq { op1: varref{id: 0} op2: 0}
} } } } }
```



Conversion

```
if (x_0 == 0)
```

# Test program generation



# Correctness testing requires encoding expectation somehow

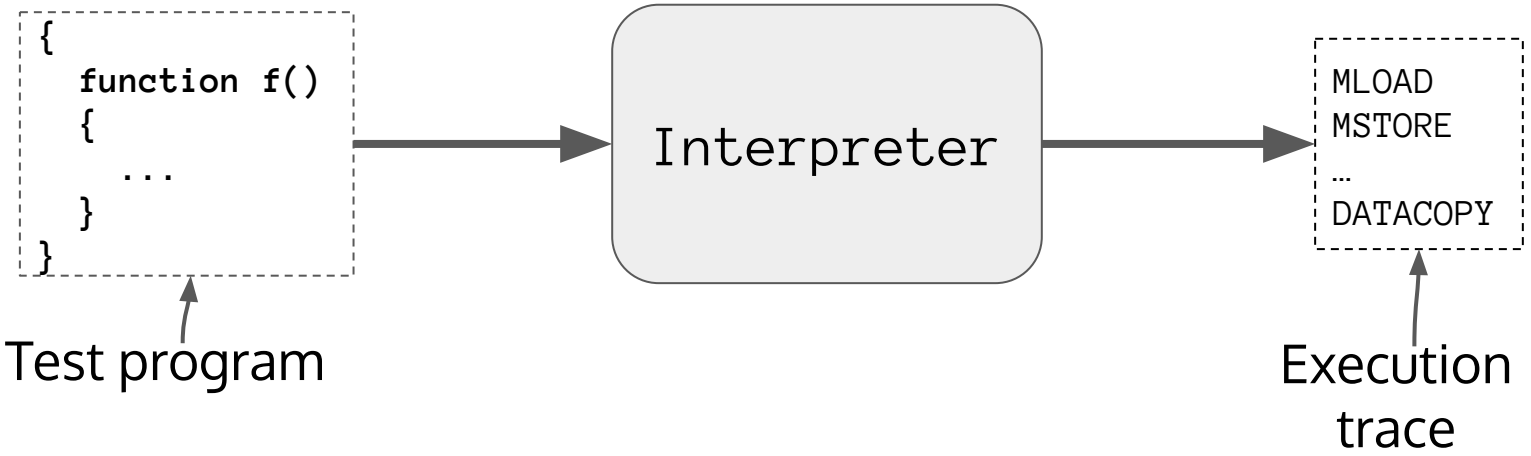
# Differential fuzzing

- Track side-effects of execution
- Run program
- Run optimized program
- Compare side-effects

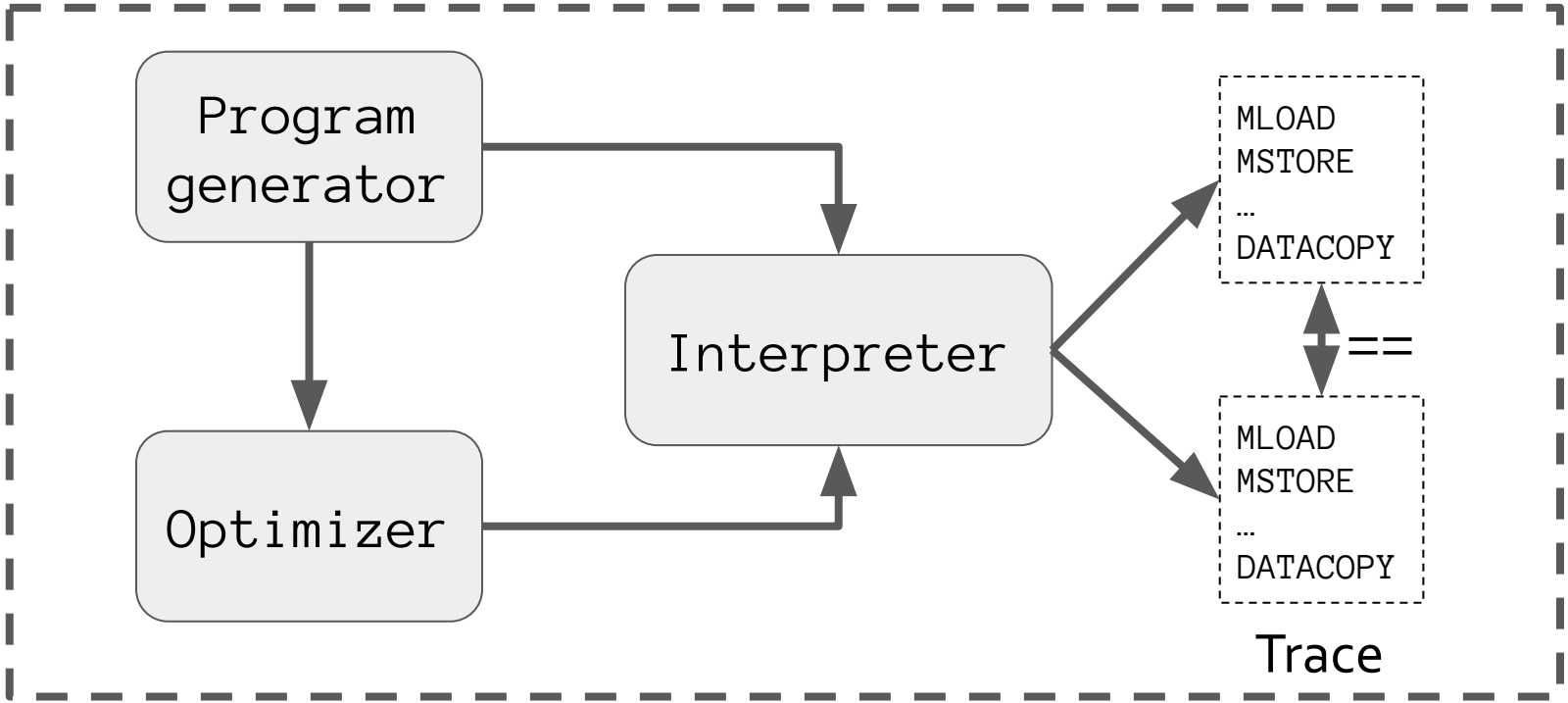
# Yul interpreter

- Interprets arbitrary yul program
- Outputs side-effects as a trace (string)

# Yul interpreter

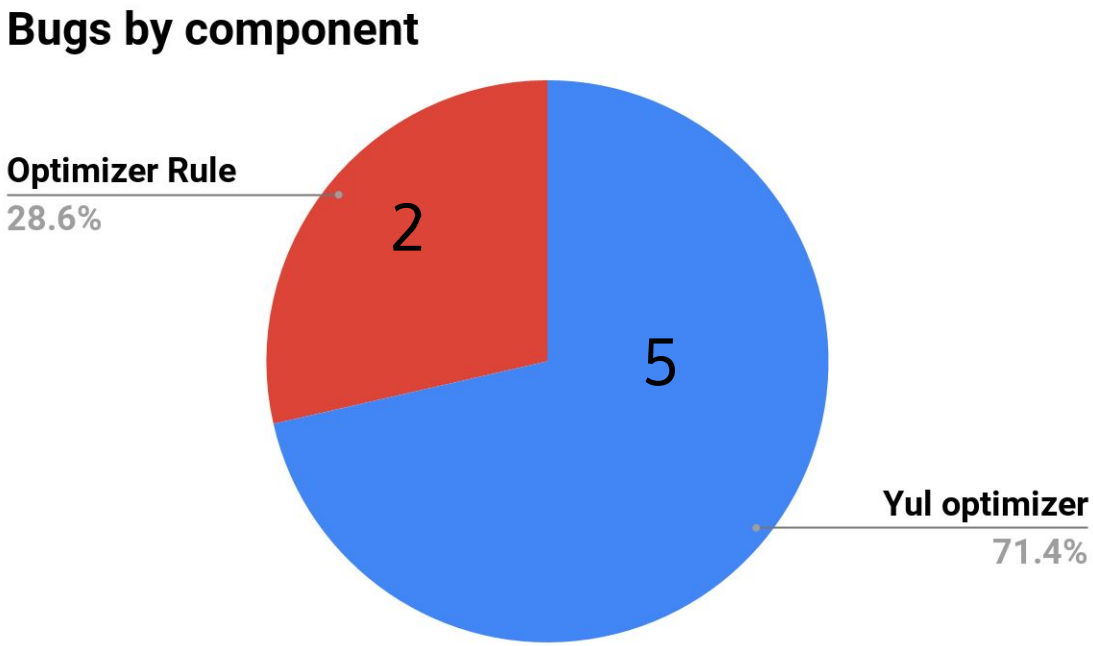


# Fuzzing Setup

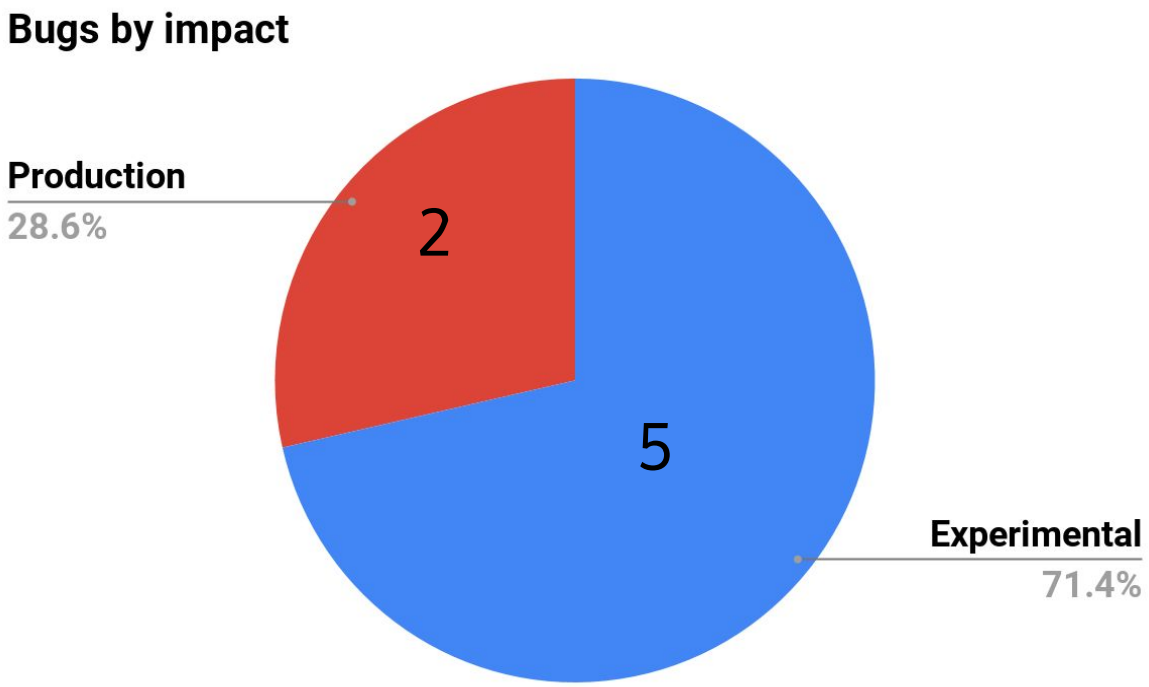


# Results

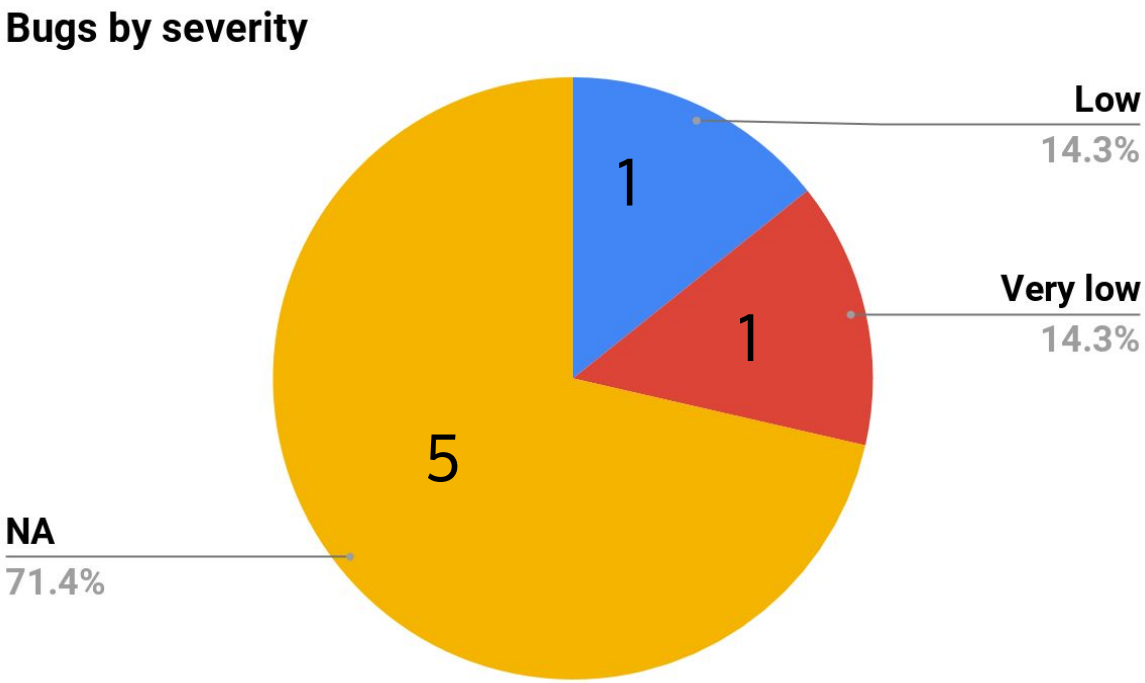
# Bugs by component



# Bugs by impact



# Bugs by severity



# Challenges

- Find high-severity bugs using fuzz testing
  - Slow test throughput (~1 test per second)
- Test Abiv2encoder
    - Generate test program (Reasonably fast)
    - Compile program (Slowest)
    - Run program on EVM (Slow)
    - Assert output validity (Very fast)

# Conclusion

# Conclusion

- Continuous structure-aware fuzzing for early bug discovery
- Useful for testing optimizer and data en/decoding
- Decent assurance
  - Evidence that it works
  - No formal guarantees though

# Thank you!

https://github.com/ethereum/solidity